Radiation-Hydrodynamics Simulation

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1 Hydrodynamics scheme

The scheme uses the principle of operator splitting, where the hydrodynamics and radiation are solved separately. The hydrodynamics is solved using a finite volume Lagrangian method (e.g. Richtmyer-Morton) and the radiation is solved using a finite difference method. The coupling between the two is done through the source terms in the hydrodynamics equations, which are updated after each radiation step.

1.1 Hydrodynamics step

The following uses the Euler equations:

$$\frac{\partial u}{\partial t} = -\frac{\partial P}{\partial m} \tag{1}$$

$$\frac{\partial V}{\partial t} = \frac{\partial u}{\partial m} \tag{2}$$

$$\frac{\partial e}{\partial t} = -P\frac{\partial u}{\partial m} \tag{3}$$

$$p = r\rho e \tag{4}$$

$$P = p + q \tag{5}$$

where E_{rad} is the energy exchange with the radiation field, which is updated after the radiation step (kept aside for now). In the lagrangian scheme, we begin with a certain pressure that applies a force on the mass elements, which causes them to accelerate. The scheme goes as follows:

1. Compute acceleration from the pressure gradient:

$$a_i^n = -2 \cdot \frac{\left(p_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n\right) - \left(p_{i-\frac{1}{2}}^n + q_{i-\frac{1}{2}}^n\right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \tag{6}$$

2. Update velocity using the acceleration:

$$u_i^{n+\frac{1}{2}} = u_i^n + a_i^n \Delta t \tag{7}$$

3. Update node positions using the updated velocities (which is exactly a helper step for updating the volume):

$$x_i^{n+1} = x_i^n + u_i^{n+\frac{1}{2}} \Delta t \tag{8}$$

4. Update the volume and density of each cell using the new node positions:

$$V_{i+\frac{1}{2}}^{n+1} = \frac{x_{i+1}^{n+1} - x_i^{n+1}}{m_{i+\frac{1}{2}}} \tag{9}$$

$$\rho_{i+\frac{1}{2}}^{n+1} = \frac{1}{V_{i+\frac{1}{2}}^{n+1}} \tag{10}$$

5. Calculate artificial viscosity q to handle shocks, using the von Neumann-Richtmyer prescription:

$$q_{i+\frac{1}{2}}^{n+1} = \begin{cases} \sigma\rho_{i+\frac{1}{2}}^{n+1}\left(u_{i+1}^{n+1} - u_i^{n+1}\right)^2 & \text{if } u_i^{n+1} > u_{i+1}^{n+1} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

6. Calculate the new spcific internal energy using the updated pressure and velocity:

$$e_{i+\frac{1}{2}}^{n+1} = \frac{e_{i+\frac{1}{2}}^n - \frac{1}{2}\left(p_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^{n+1}\right)\left(\frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}}\right)}{1 + \frac{\gamma-1}{2}\rho_{i+\frac{1}{2}}^{n+1}\frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}}} \tag{12}$$

7. Calculate the new pressure using the equation of state:

$$p_{i+\frac{1}{2}}^{n+1} = (\gamma - 1)\rho_{i+\frac{1}{2}}^{n+1} e_{i+\frac{1}{2}}^{n+1} \tag{13}$$

8. Compute acceleration from the **new** pressure gradient:

$$a_i^{n+1} = -2 \cdot \frac{\left(p_{i+\frac{1}{2}}^{n+1} + q_{i+\frac{1}{2}}^{n+1}\right) - \left(p_{i-\frac{1}{2}}^{n+1} + q_{i-\frac{1}{2}}^{n+1}\right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \tag{14}$$

9. Update velocity using the new acceleration:

$$u_i^{n+1} = u_i^{n+\frac{1}{2}} + a_i^{n+1}\frac{\Delta t}{2} \tag{15}$$

and go back to step 2 for the next iteration. Make sure to understand the reason for the half time step in the velocity update - it makes the scheme second-order and less noisy.

1.2 Coupling with radiation

This is all good for the hydrodynamics step, but we need to couple it with radiation! After step 9, we define:

$$e^* = e^{n+1} \tag{16}$$

which means that the energy isn't updated yet! we then assume that $U_m^n = \rho e^*$ is the energy density of the matter.

2 Radiation Scheme

2.1 Physical PDEs

1. Writing the diffusion equation for the radiation energy $E = aT_r^4$ and the effective material radiative energy $U_R = aT_m^4$ (the energy that the material, modeled as a black body, would radiate outwards):

$$\frac{\partial E}{\partial t} + \rho\frac{\partial F}{\partial m} = \chi c\sigma(U_R - E) \tag{17}$$

$$\frac{\partial U_m}{\partial t} = -\chi c\sigma(U_R - E) \Rightarrow \frac{1}{\beta}\frac{\partial U_R}{\partial t} = -\chi c\sigma(U_R - E) \tag{18}$$

$$F = -D\rho\frac{\partial E}{\partial m} \quad \left(D = \frac{c}{3\sigma(T,\rho)}\right) \tag{19}$$

In the equation here, one assumes that during the radiation substep we keep ρ^* (to be defined later) fixed, and then linearize $U_R(U_m)$ locally around U_m^* to get the relation $\frac{1}{\beta}\frac{\partial U_R}{\partial t} = -\chi c\sigma(U_R - E)$, where $\beta = \frac{dU_R}{dU_m}|_{U_m^*}$ is the coupling coefficient between the radiation and matter that satifies $dU_R \approx \beta dU_m$ around U_m^*.

2. The self similar relations for the opacity (and thus the diffusion constant):

$$\frac{1}{\kappa_R} = gT^{\alpha}\rho^{-\lambda} \tag{20}$$

$$\frac{1}{\sigma(T,\rho)} = \frac{1}{\rho\kappa_R(T)} = gT^{\alpha}\rho^{-\lambda-1} \Rightarrow D = \frac{c}{3}gT^{\alpha}\rho^{-\lambda-1} \tag{21}$$

and the self similar relation for the internal energy:

$$e = fT^{\gamma}\rho^{-\mu} \tag{22}$$

where e is the specific internal energy, which means the material energy density is:

$$U_m(T) = \rho e = fT^{\gamma}\rho^{-\mu+1} \tag{23}$$

3. Notice that we've converted the space derivatives to mass derivatives, using the relation:

$$\frac{\partial}{\partial x} = \frac{\partial m}{\partial x}\frac{\partial}{\partial m} = \rho\frac{\partial}{\partial m} \tag{24}$$

as done in the Appendix.

4. To sum up, we have **a system of PDEs over T_m and T_R**:

$$\frac{\partial\left(aT_r^4\right)}{\partial t} - \rho\frac{\partial}{\partial m}\left(D\rho\frac{\partial(aT_r^4)}{\partial m}\right) = \chi c\sigma a(T_m^4 - T_r^4) \tag{25}$$

$$\frac{1}{\beta}\frac{\partial\left(aT_m^4\right)}{\partial t} = -\chi c\sigma a(T_m^4 - T_r^4) \tag{26}$$

The system is solved over a domain of N cells with BC of ... to be defined.

2.2 Finite Difference Scheme

2.2.1 Discretization of a lagrangian PDE

1. In the diffusion equation, we have terms like:

$$\frac{\partial E}{\partial x} = \rho\frac{\partial E}{\partial m} \tag{27}$$

but how are they being discretized? We can discretize them by simply writing (leaving ρ outside of discussion for now):

$$\frac{\partial E}{\partial m} \approx \frac{E^n_{i+\frac{1}{2}} - E^n_{i-\frac{1}{2}}}{\Delta m_i} \tag{28}$$

$$\Delta m_i := \frac{1}{2}\left((m_{i+1} - m_i) + (m_i - m_{i-1})\right) = \frac{1}{2}(\Delta m_{i+\frac{1}{2}} + \Delta m_{i-\frac{1}{2}}) \quad (29)$$

where the lagrangina coordinate m_i for $i = 0, ..., N$ is defined as the mass of the material up to the vertex located at x_i. Therefore $m_i = \sum_{j=0}^{i} \Delta m_{i+\frac{1}{2}}$ where $\Delta m_{i+\frac{1}{2}} = m_{i+1} - m_i$ is the mass of the cell $j = i + \frac{1}{2}$ $(j = 1, ..., N)$ between the vertices i and $i + 1$.

2. It may also be denoted $\Delta m_i = m_{i+\frac{1}{2}} - m_{i-\frac{1}{2}}$ where $m_{i+\frac{1}{2}} = m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}}$ and $m_{i-\frac{1}{2}} = m_{i-1} + \frac{1}{2}\Delta m_{i-\frac{1}{2}}$.

$$\begin{aligned}\Delta m_i &= m_{i+\frac{1}{2}} - m_{i-\frac{1}{2}} \\ &= \left(m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}}\right) - \left(m_{i-1} + \frac{1}{2}\Delta m_{i-\frac{1}{2}}\right) = \frac{1}{2}\left(\Delta m_{i+\frac{1}{2}} + \Delta m_{i-\frac{1}{2}}\right)\end{aligned} \quad (30)$$

3. One should also regard the problem when taking the second derivative of E with respect to m - rising from the first derivative of the flux!

$$\frac{\partial F}{\partial x} = \rho\frac{\partial F}{\partial m} = \rho\frac{\partial}{\partial m}\left(-D\rho\frac{\partial E}{\partial m}\right) \quad (31)$$

What a mess! How are we supposed to discretize this? The solution is to discretize step by step - first the discretize the first derivative of the flux, and then the first derivative of each E term.

$$\frac{\partial F}{\partial m} \approx \frac{F^n_{i+1} - F^n_i}{\Delta m_{i+\frac{1}{2}}} \quad (32)$$

and it makes sense because the flux is defined at the faces of the cells $i + \frac{1}{2}$ whose mass is actually $\Delta m_{i+\frac{1}{2}} = m_{i+1} - m_i$. Then, one should discretize each of the flux terms, as we have already done for the first derivative of E:

$$\begin{aligned}F^n_{i+1} &= -\rho^n_{i+1}D^n_{i+1}\frac{E^n_{i+\frac{3}{2}} - E^n_{i+\frac{1}{2}}}{\Delta m_{i+1}} \\ F^n_i &= -\rho^n_i D^n_i\frac{E^n_{i+\frac{1}{2}} - E^n_{i-\frac{1}{2}}}{\Delta m_i}\end{aligned} \quad (33)$$

where Δm_i, ρ_i and D_i terms are all explained in the previous bullets.

4. NOTE that in our case $\Delta m_{i+\frac{1}{2}}$ is constant for all i and therefore $\Delta m_i = const$ and is a **predefined parameter** of the simulation.

2.2.2 Discretization of D and ρ faces

Since the derivative is a property of the vertex located between two adjacent cells, we should also pay attention to the fact that F, D and ρ should be defined at the faces of the cells (e.g. the vertices).

Since the natural definition of D and ρ is at the cell centers (we are used to terms like $\rho^n_{i+\frac{1}{2}}$ and $D^n_{i+\frac{1}{2}}$), we need to define them at the faces of the cells using some kind of averaging.

5. The most common way to average D and ρ at the faces is to use the average mean:

$$\begin{aligned} D_{i+\frac{1}{2}} &= \frac{D_i + D_{i+1}}{2} \\ \rho_{i+\frac{1}{2}} &= \frac{\rho_i + \rho_{i+1}}{2} \end{aligned} \tag{34}$$

6. In our simulation, the constant `HARMONIC_MEAN` is set by default to `False`, which means that we use the arithmetic mean to average D and ρ at the faces.

2.2.3 Solution of the discretized system of equations

7. We've seen that one can **eliminate the effective radiation density** U_R from the finite-difference equation for the matter energy. The discretized equation for the material energy density is:

$$\frac{1}{\beta^n_{i+\frac{1}{2}}} \frac{[U_R]^{n+1}_{i+\frac{1}{2}} - [U_R]^n_{i+\frac{1}{2}}}{\Delta t} = \chi c \sigma^n_{i+\frac{1}{2}} (E^{n+1}_{i+\frac{1}{2}} - [U_R]^{n+1}_{i+\frac{1}{2}}) \tag{35}$$

$$(1 + \beta^n_{i+\frac{1}{2}} \Delta t \chi c \sigma^n_{i+\frac{1}{2}})[U_R]^{n+1}_{i+\frac{1}{2}} = \beta^n_{i+\frac{1}{2}} \Delta t \chi c \sigma^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{1}{2}} + [U_R]^n_{i+\frac{1}{2}} \tag{36}$$

We want to solve for $[U_R]^{n+1}_{i+\frac{1}{2}}$. First, we can denote $A^n_{i+\frac{1}{2}} = \beta^n_{i+\frac{1}{2}} \Delta t \chi c \sigma^n_{i+\frac{1}{2}}$ as the coupling coefficient between the radiation and matter, which gives us:

$$(1 + A^n_{i+\frac{1}{2}})[U_R]^{n+1}_{i+\frac{1}{2}} = A^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{1}{2}} + [U_R]^n_{i+\frac{1}{2}} \tag{37}$$

$$\boxed{[U_R]^{n+1}_{i+\frac{1}{2}} = \frac{A^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} + \frac{1}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}}} \tag{38}$$

8. Furthermore, after subtituting the expressions for $[U_R]^{n+1}_{i+\frac{1}{2}}$ & F^n_i into the discretized equation for the radiation energy E, we get a tridiagonal implicit scheme for E^{n+1}_i:

$$\boxed{a^n_{i+\frac{1}{2}} E^{n+1}_{i-\frac{1}{2}} + b^n_{i+\frac{1}{2}} E^n_{i+\frac{1}{2}} + c^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{3}{2}} = d^n_{i+\frac{1}{2}}} \tag{39}$$

where the coefficients $a^n_i, b^n_i, c^n_i, d^n_i$ are calculated in the appendix.

9. We notice that the coefficients $a^n_i, b^n_i, c^n_i, d^n_i$ depend on the hydrodynamics variables (e.g. density, temperature) through the diffusion constant D and the opacity σ. The dependency is given through Rosen's formulae for the opacity and the internal energy. First, one should find the temperature at the previous time step:

$$T^n_{i+\frac{1}{2}} = \sqrt[4]{\frac{[U_R]^n_{i+\frac{1}{2}}}{a}} \tag{40}$$

and then substitute it into the relations for $D(T), \beta(T)$:

$$D^n_{i+\frac{1}{2}} = \frac{c}{3} g \left[T^n_{i+\frac{1}{2}}\right]^{\alpha} \left[\rho^n_{i+\frac{1}{2}}\right]^{-\lambda-1} \tag{41}$$

$$\beta^n_{i+\frac{1}{2}} = \frac{dU_R}{dU_m} = \frac{4a}{f\gamma} \left[T^n_{i+\frac{1}{2}}\right]^{4-\gamma} \left[\rho^n_{i+\frac{1}{2}}\right]^{\mu-1} \tag{42}$$

$$\frac{1}{\sigma^n_{i+\frac{1}{2}}} = \frac{1}{\rho\kappa_R(T)} = g \left[T^n_{i+\frac{1}{2}}\right]^{\alpha} \left[\rho^n_{i+\frac{1}{2}}\right]^{-\lambda-1} \tag{43}$$

NOTE: In the coupled scheme, we will use the temperature $T^*_{i+\frac{1}{2}}$ computed from the hydrodynamics step **directly** to calculate D, β, σ and substitute in the tridiagonal system, no need to go through U_R in the coupled scheme.

10. To sum up, the complete algorithm for the radiation scheme is as follows:

    (a) Compute the temperature at the previous time step using the effective radiation density U_R.

    (b) Compute the diffusion constant D, the coupling coefficient β and the opacity σ using the temperature and density at the previous time step.

    (c) Compute the coefficients $a^n_i, b^n_i, c^n_i, d^n_i$ for the tridiagonal system.

(d) Solve the tridiagonal system for E_i^{n+1} using the Thomas algorithm.

(e) Update the effective radiation density U_R^{n+1} using the expression derived above.

3 Coupling the two schemes

1. We first perform the hydrodynamics step, which gives us the updated density ρ^{n+1} and specific internal energy e^{n+1}. We call it e^* because we haven't updated the energy with the radiation source term yet.

2. In the radiation scheme, the self-similar profiles for the opacity and internal energy depend on the material temperature. Since we know that $U_m = \rho e^*$, we can compute the temperature at the previous time step using the relation:

$$e^*_{i+\frac{1}{2}} = f\left[T^*_{i+\frac{1}{2}}\right]^{\gamma}\left[\rho^*_{i+\frac{1}{2}}\right]^{-\mu} \Rightarrow T^*_{i+\frac{1}{2}} = \left(\frac{e^*_{i+\frac{1}{2}}}{f}\left[\rho^*_{i+\frac{1}{2}}\right]^{\mu}\right)^{\frac{1}{\gamma}} \tag{44}$$

3. Apply the radiation scheme using the temperature and density at the previous time step to compute β, D and σ which give us the coefficients for the tridiagonal system, which we solve to get $E^{n+1}_{i+\frac{1}{2}}$ and $[U_R]^{n+1}_{i+\frac{1}{2}}$.
NOTE: The chat proposes to perform a Picard loop, where we update the temperature using the radiation energy until convergence.

4. We then update the specific internal energy after the second source term of radiation is applied:

$$e^{n+1}_{i+\frac{1}{2}} = \frac{[U_m]^{n+1}_{i+\frac{1}{2}}}{\rho^{n+1}_{i+\frac{1}{2}}} \tag{45}$$

5. Finally, we update the hydrodynamic pressure using the equation of state:

$$p^{n+1}_{i+\frac{1}{2}} = r\rho^{n+1}_{i+\frac{1}{2}}e^{n+1}_{i+\frac{1}{2}} \tag{46}$$

and go back to the hydrodynamics step for the next iteration!

4 Corrections

4.1 Correcting the scheme with black physics

1. In the radiation diffusion equation, the finite-difference form that we reached previously is:

$$LHS = \frac{\partial E}{\partial t} + \rho \frac{\partial F}{\partial m} \tag{47}$$

$$\Updownarrow \tag{48}$$

$$\begin{aligned} LHS = \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{m_{i+\frac{3}{2}}} \right)}^{-F^{n+1}_{i+1}} \\ + \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{m_{i+\frac{1}{2}}} \right)}^{-F^{n+1}_{i}} \end{aligned} \tag{49}$$

but wait! If we assume black physics, the radiation energy should be changed to E^* and not E^{n+1}. That's why the new LHS is:

$$\begin{aligned} LHS = \frac{\color{blue}{E^{*}_{i+\frac{1}{2}}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i+\frac{3}{2}} \frac{2D^n_{i+\frac{1}{2}} D^n_{i+\frac{3}{2}}}{D^n_{i+\frac{1}{2}} + D^n_{i+\frac{3}{2}}} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{m_{i+\frac{3}{2}}} \right)}^{-F^{n+1}_{i+1}} \\ + \frac{\rho^*_{i+\frac{1}{2}}}{m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i-\frac{1}{2}} \frac{2D^n_{i-\frac{1}{2}} D^n_{i+\frac{1}{2}}}{D^n_{i-\frac{1}{2}} + D^n_{i+\frac{1}{2}}} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{m_{i+\frac{1}{2}}} \right)}^{-F^{n+1}_{i}} \end{aligned} \tag{50}$$

2. And the RHS remains implicit, so it's the same as before:

$$RHS = \chi c \sigma (U_R - E) \tag{51}$$

$$\Updownarrow \tag{52}$$

$$
\begin{aligned}
RHS &= \chi c\sigma^n_{i+\frac{1}{2}} \left(\frac{A^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} + \frac{1}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - E^{n+1}_{i+\frac{1}{2}} \right) \\
&= \chi c\sigma^n_{i+\frac{1}{2}} \left(\frac{1}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{1}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} \right) \\
&= \frac{\chi c\sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{\chi c\sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}}
\end{aligned}
\tag{53}
$$

3. Conclusion: The new LHS and RHS define a new tridiagonal system of equations for the radiation energy density E^*, which creates a new sparse matrix that needs to be solved. Ask Shay about this conclusion - if it's correct, it requires a new algebric derivation of the tridiagonal constants, and implementation in python.

4. CORRECTION: My conclusion is incorrect. E^n should be replaced by E^* and NOT E^{n+1}. It makes sense because we suppose that the radiation energy should also be changed.

5. Implementation: The only coefficient that had previously (prior to the correction) included E^n is d^n. The new formula for d^n is:

$$
d^n_i = \frac{\chi c\sigma^n_{i+\frac{1}{2}}}{1+A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} + \frac{1}{\Delta t} E^*_{i+\frac{1}{2}} \tag{54}
$$

or in language of arrays:

$$
d^n_j = \frac{\chi c\sigma^n_j}{1+A^n_j} [U_R]^n_j + \frac{1}{\Delta t} E^*_j \tag{55}
$$

4.2 Black (LTE) radiation scheme

1. One can also solve the rad-hydro scheme under the LTE assumption that leads to the following simpler equation:

$$
\begin{aligned}
\frac{\partial E}{\partial t} + \frac{\partial F}{\partial x} &= Q \\
C_V \frac{\partial T}{\partial t} &= -Q
\end{aligned}
\tag{56}
$$

where we denoted:

$$
\begin{aligned}
Q &= \Sigma'_a (aT^4_m - E) \\
F &= -D\frac{\partial E}{\partial x}
\end{aligned}
\tag{57}
$$

$$\Downarrow \tag{58}$$

$$\begin{aligned} \frac{\partial e}{\partial t} + \frac{\partial F}{\partial x} &= 0 \\ F &= -D\frac{\partial e}{\partial x} \end{aligned} \tag{59}$$

The LTE assumption means that $T_m = T_R = T$ and the relation between the energy densities and the temperatures is given by:

$$\begin{aligned} e &= fT^{\beta}\rho^{-\mu} \quad \text{Rosen's notation} \\ E &= aT^4 \quad \text{Stefan-Boltzmann law} \end{aligned} \tag{60}$$

2. The finite-difference schem for this equation is: The starting point is the coupled energy equation in LTE conditions from above:

$$\begin{aligned} \frac{\partial e}{\partial t} &= \frac{\partial}{\partial x}\left(D\frac{\partial e}{\partial x}\right) \\ \frac{\partial e}{\partial t} &= \rho\frac{\partial}{\partial m}\left(D\rho\frac{\partial e}{\partial m}\right) \end{aligned} \tag{61}$$

In the Lagrangian framework used here, the cell centers are indexed by $i+\frac{1}{2}$, while the cell faces (vertices) are indexed by i. In Lagrangian coordinates, the spatial derivative is written in terms of mass coordinates, using $1/\Delta x = \rho^*/\Delta m$, where Δm is the cell mass and ρ^* is the (hydro-updated) density at the cell center.

The finite-difference (implicit) scheme for the diffusion equation in Lagrangian coordinates reads:

$$\begin{aligned} LHS &= \frac{e^{n+1}_{i+\frac{1}{2}} - e^{n}_{i+\frac{1}{2}}}{\Delta t} \\ RHS &= \frac{1}{\Delta m_{i+\frac{1}{2}}}\left[D_{i+1}\rho^*_{i+1}\frac{e^{n+1}_{i+\frac{3}{2}} - e^{n+1}_{i+\frac{1}{2}}}{\Delta m_{i+1}} - D_i\rho^*_i\frac{e^{n+1}_{i+\frac{1}{2}} - e^{n+1}_{i-\frac{1}{2}}}{\Delta m_i}\right] \end{aligned} \tag{62}$$

where D_i is the diffusion coefficient at face (vertex) i, Δm_i is the mass of the cell between vertices $i-1$ and i, and $\rho^*_{i+\frac{1}{2}}$ is the density at the cell center after the hydro update.

The harmonic mean is still used for D_i:

$$D_i = \frac{2D_{i-\frac{1}{2}}D_{i+\frac{1}{2}}}{D_{i-\frac{1}{2}} + D_{i+\frac{1}{2}}} \tag{63}$$

(harmonic mean), and similarly for D_{i+1}.

Rearranging terms to bring all unknowns $e^{n+1}_{j+\frac{1}{2}}$ to the left yields the tridiagonal system:

$$\boxed{a_{i+\frac{1}{2}}\, e^{n+1}_{i-\frac{1}{2}} + b_{i+\frac{1}{2}}\, e^{n+1}_{i+\frac{1}{2}} + c_{i+\frac{1}{2}}\, e^{n+1}_{i+\frac{3}{2}} = e^{n}_{i+\frac{1}{2}}} \tag{64}$$

where the boxed coefficients are:

$$\boxed{a_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_i}\frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} D_i} \tag{65}$$

$$\boxed{b_{i+\frac{1}{2}} = 1 + \frac{\Delta t}{\Delta m_{i+\frac{1}{2}}}\frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} (D_i + D_{i+1})} \tag{66}$$

$$\boxed{c_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_{i+1}}\frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} D_{i+1}} \tag{67}$$

Here, the index $i + \frac{1}{2}$ denotes the cell center, the left neighbor is at $i - \frac{1}{2}$ and the right neighbor at $i + \frac{3}{2}$, consistent with the Lagrangian grid convention.

For practical implementation, let $j = i + \frac{1}{2}$ and define arrays for a_j, b_j, c_j, e_j, etc. The resulting tridiagonal system is:

$$a_j\, e^{n+1}_{j-1} + b_j\, e^{n+1}_{j} + c_j\, e^{n+1}_{j+1} = e^{n}_{j} \tag{68}$$

which can be efficiently solved using the Thomas algorithm or any suitable tridiagonal matrix solver.

Here, $e^{n+1}_{i+\frac{1}{2}}$ is the energy at cell center $i + \frac{1}{2}$ at the new timestep, and the resulting system is tridiagonal and can be solved by the Thomas algorithm.

Note: If e and E are directly connected via LTE (e.g. $e = fT^\beta \rho^{-\mu}$ and $E = aT^4$) and the same equation holds for E, a similar discrete scheme applies to E.

5 Appendix: Derivation of the tridiagonal form coefficients in Lagrangian coordinates

1. The discretized equation for the radiation energy in Lagrangian coordinates is:

$$\frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} + \rho^*_{i+\frac{1}{2}} \frac{F^{n+1}_{i+1} - F^{n+1}_{i}}{\Delta m_{i+\frac{1}{2}}} = \chi c \sigma^n_{i+\frac{1}{2}} ([U_R]^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i+\frac{1}{2}}) \quad (69)$$

2. Recalling that the flux and U_R are given as expressions of E, we can substitute them into the equation above to get a tridiagonal system for $E^{n+1}_{i+\frac{1}{2}}$.

   <u>LHS:</u> The flux is given by:

$$F^n_i = -\rho^*_i D^n_i \frac{E^n_{i+\frac{1}{2}} - E^n_{i-\frac{1}{2}}}{\Delta m_i} \quad (70)$$

   NOTE: The radiation energy density E is taken implicitely, and the D face is taken exaplicitly - because of numeric considerations (I need to ask Shay about this). Substituting the fluxes into the balance equation gives:

$$LHS = \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left(\rho^*_{i+1} D^n_{i+1} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{\Delta m_{i+1}} \right)}^{-F^{n+1}_{i+1}} + \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left(\rho^*_i D^n_i \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{\Delta m_i} \right)}^{-F^{n+1}_{i}} \quad (71)$$

Rearranging the terms gives us the tridiagonal form:

$$
\begin{aligned}
LHS = &\left(-\frac{\rho^*_{i+\frac{1}{2}}\rho^*_i D^n_i}{\Delta m_{i+\frac{1}{2}}\Delta m_i}\right) E^{n+1}_{i-\frac{1}{2}} \\
&+ \left[\frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}}\left(\frac{\rho^*_i D^n_i}{\Delta m_i} + \frac{\rho^*_{i+1} D^n_{i+1}}{\Delta m_{i+1}}\right) + \frac{1}{\Delta t}\right] E^{n+1}_{i+\frac{1}{2}} \\
&- \left(\frac{\rho^*_{i+\frac{1}{2}}\rho^*_{i+1} D^n_{i+1}}{\Delta m_{i+\frac{1}{2}}\Delta m_{i+1}}\right) E^{n+1}_{i+\frac{3}{2}} \\
&- \frac{1}{\Delta t} E^n_{i+\frac{1}{2}}
\end{aligned}
\tag{72}
$$

<u>RHS:</u> Substituting the expression for $[U_R]^{n+1}_{i+\frac{1}{2}}$ gives:

$$
\begin{aligned}
RHS &= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{A^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} + \frac{1}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - E^{n+1}_{i+\frac{1}{2}}\right) \\
&= \chi c \sigma^n_{i+\frac{1}{2}} \left(\frac{1}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{1}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}}\right) \\
&= \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}}
\end{aligned}
\tag{73}
$$

3. Finally, rearranging the equation $LHS = RHS$ gives us the tridiagonal form:

$$
a^n_{i+\frac{1}{2}} E^{n+1}_{i-\frac{1}{2}} + b^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{1}{2}} + c^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{3}{2}} = d^n_{i+\frac{1}{2}} \tag{74}
$$

and the coefficients are given by:

$$\boxed{a^n_{i+\frac{1}{2}} = -\frac{\rho^*_{i+\frac{1}{2}}\rho^*_i D^n_i}{\Delta m_{i+\frac{1}{2}}\Delta m_i}} \tag{75}$$

$$\boxed{b^n_{i+\frac{1}{2}} = \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}}\left(\frac{\rho^*_i D^n_i}{\Delta m_i} + \frac{\rho^*_{i+1} D^n_{i+1}}{\Delta m_{i+1}}\right) + \frac{1}{\Delta t} + \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}}} \tag{76}$$

$$\boxed{c^n_{i+\frac{1}{2}} = -\frac{\rho^*_{i+\frac{1}{2}}\rho^*_{i+1} D^n_{i+1}}{\Delta m_{i+\frac{1}{2}}\Delta m_{i+1}}} \tag{77}$$

$$\boxed{d^n_{i+\frac{1}{2}} = \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}}[U_R]^n_{i+\frac{1}{2}} + \frac{1}{\Delta t}E^n_{i+\frac{1}{2}}} \tag{78}$$

where $A^n_{i+\frac{1}{2}} = \beta^n_{i+\frac{1}{2}}\Delta t \chi c \sigma^n_{i+\frac{1}{2}}$ is the coupling coefficient between the radiation and matter.

Two things to understand

I need to understand how the Flek & Cummings coefficient f is related to the coupling coefficient A.

5.1 Deeper understanding of the tridiagonal form

1. First, let's make order in the celebration of indexes. Our staggered grid is as follows:

Start BC		**Internal**	...	**Internal**		**End BC**
$i = 0$ $j = \frac{1}{2}$	$i = \frac{1}{2}$ $\boxed{j = 1}$	$i = 1$ $j = \frac{3}{2}$	... ...	$i = N-1$ $j = N - \frac{1}{2}$	$i = N - \frac{1}{2}$ $\boxed{j = N}$	$i = N$ $j = N + \frac{1}{2}$
node ρ_0, D_0	cell center $E_0, [U_R]_0$ $\rho_{1/2}, D_{1/2}$ $\beta_{1/2}, \sigma_{1/2}$	node ρ_1, D_1	... ...	node ρ_{N-1}, D_{N-1}	cell center $E_N, [U_R]_N$ $\rho_{N-1/2}, D_{N-1/2}$ $\beta_{N-1/2}, \sigma_{N-1/2}$	node ρ_N, D_N

where we defined $j = i + \frac{1}{2}$ as the index for the cell centers. Now, the indexes values are:

$$\begin{aligned} i &= 0, 1, 2, ..., N \quad \text{(nodes)} \\ j &= 1, 2, ..., N \quad \text{(cell centers)} \end{aligned} \tag{79}$$

2. We wish to write the tridiagonal equations in a comfortable matrix form. Under the assumption that the radiation energy array E_j is defined on cell centers $j = 1, ..., N$ but also at $j = 0$ and $j = N + 1$ (ghost cells in which the value is defined via the BC), we get the follows.
For $j = 1, ..., N$ where $i = j - \frac{1}{2}$:

$$\begin{gathered} a^n_{i+\frac{1}{2}} E^{n+1}_{i-\frac{1}{2}} + b^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{1}{2}} + c^n_{i+\frac{1}{2}} E^{n+1}_{i+\frac{3}{2}} = d^n_{i+\frac{1}{2}} \\ \Downarrow \\ a^n_j E^{n+1}_{j-1} + b^n_j E^{n+1}_j + c^n_j E^{n+1}_{j+1} = d^n_j \end{gathered} \tag{80}$$

which build the following matrix system of equations:

$$\begin{bmatrix} a_1 & b_1 & c_1 & 0 & \cdots & 0 & 0 & \\ 0 & a_2 & b_2 & c_2 & \cdots & 0 & 0 & \\ 0 & 0 & a_3 & b_3 & c_3 & \cdots & 0 & \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \\ 0 & \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & a_N & b_N & c_N \end{bmatrix} \begin{bmatrix} E^{n+1}_0 \\ E^{n+1}_1 \\ \vdots \\ \vdots \\ \vdots \\ E^{n+1}_N \\ E^{n+1}_{N+1} \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \tag{81}$$

which gives us a system of N **equations with** $N + 2$ **variables**.

Before we discuss the definition of the cell-centered coefficients $a^n_j, b^n_j, c^n_j, d^n_j$ for $j = 1, ..., N$, let's think of how one would implement them on a **computer program**.

Implementation of the coefficient calculations

There are two important factors to consider when calculating the coefficients:

(a) How one correctly implement the cell-valued arrays and the face-valued arrays in a computer program, where there isn't a way to create half indexes or differentiate between i and j in a simple array.

(b) The validity of the indexes of the coefficients and other variables are affected by the boundary conditions.

5.1.1 The matrix equation for different BCs

(a) In **Dirichlet BC** we define:

$$E_0^n = T_s(t) \tag{82}$$
$$E_{N+1}^n = 0 \tag{83}$$

so that the system of equations is now:

$$\begin{bmatrix} a_1 & b_1 & c_1 & 0 & \cdots & 0 & 0 & \\ 0 & a_2 & b_2 & c_2 & \cdots & 0 & 0 & \\ 0 & 0 & a_3 & b_3 & c_3 & \cdots & 0 & \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \\ 0 & \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & a_N & b_N & c_N \end{bmatrix} \begin{bmatrix} T_s(t) \\ E_1^{n+1} \\ \vdots \\ \vdots \\ \vdots \\ E_N^{n+1} \\ 0 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \tag{84}$$

And now we are left with N equations with N unknowns! (in fact we have just added the two missing equations, that directly solve for E_0 and E_{N+1}).

At this point, the 1st and N'th equations become:

$$a_1 T_s(t) + b_1 E_1^{n+1} + c_1 E_2^{n+1} = d_1 \tag{85}$$
$$a_N E_{N-1}^{n+1} + b_N E_N^{n+1} + c_N \cdot 0 = d_N \tag{86}$$

which can be written as:

$$b_1 E_1^{n+1} + c_1 E_2^{n+1} = d_1 - a_1 T_s(t) \tag{87}$$
$$a_N E_{N-1}^{n+1} + b_N E_N^{n+1} = d_N \tag{88}$$

So that the elements a_1 and c_N are ommitted, and their effect is inserted inside the d coefficients to provide the **reduced matrix equation:**

$$\begin{bmatrix} b_1 & c_1 & 0 & \cdots & 0 & \\ a_2 & b_2 & c_2 & \cdots & 0 & \\ 0 & a_3 & b_3 & c_3 & \cdots & \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & 0 & \cdots & 0 & a_N & b_N \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ \vdots \\ \vdots \\ \vdots \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} d_1 - a_1 T_s(t) \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \tag{89}$$

(b) In **Marshak BC** we define:

$$F_0^n = F(T_{bath}) \tag{90}$$
$$E_{N+1}^n = 0 \tag{91}$$

Through a derivation that appears in detail in Appendix 6 one can reduce the equation at the surface ($i = 0$) into an equation of the form:

$$\tilde{b}_1 E_1^{n+1} + \tilde{c}_1 E_2^{n+1} = \tilde{d}_1 \tag{92}$$

where $\tilde{b}_1, \tilde{c}_1, \tilde{d}_1$ are of course defined in Appendix 6. Because this new equation completely replaces the standard diffusion equation for the first cell, the ghost cell E_0^{n+1} is algebraically eliminated from our active unknowns. Therefore, the system remains exactly N equations for the N internal cells:

$$\begin{bmatrix} \tilde{b}_1 & \tilde{c}_1 & 0 & \cdots & 0 & 0 \\ a_2 & b_2 & c_2 & \cdots & 0 & 0 \\ 0 & a_3 & b_3 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & c_{N-2} & 0 \\ 0 & \cdots & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & 0 & \cdots & 0 & a_N & b_N \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ E_2^{n+1} \\ E_3^{n+1} \\ \vdots \\ E_{N-1}^{n+1} \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} \tilde{d}_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \tag{93}$$

We are now left with exactly N equations in N unknowns, which perfectly matches the implementation size in the numerical code.

5.1.2 The tridiagonal coefficient calculation

$\rho, \Delta m, D, \sigma$ are all given as arrays for the cell centers (i.e. $\rho_j =$ `rho[j - 1]`) for $j = 1, ..., N$. Starting from this principle, we can calculate the face-valued arrays as follows.
<u>For $i = 0, ..., N$</u> (notice the `[1:], [:-1]` notation that forbids invalid indices):

$$\begin{aligned} \rho_i^* &= \text{avg}(\rho_{i-\frac{1}{2}}^*, \rho_{i+\frac{1}{2}}^*) \implies \texttt{rho\_face[:]=avg(rho[:-1], rho[1:])} \\ D_i^n &= \text{avg}(D_{i-\frac{1}{2}}^n, D_{i+\frac{1}{2}}^n) \implies \texttt{D\_face[:]=avg(D[:-1], D[1:])} \\ \Delta m_i &= \text{avg}(\Delta m_{i-\frac{1}{2}}, \Delta m_{i+\frac{1}{2}}) \implies \texttt{m\_face[:]=avg(m\_cells[:-1], m\_cells[1:])} \end{aligned} \tag{94}$$

of `size=N-1`. We will define:

$$f_i^* = \frac{\rho_i^* D_i^n}{\Delta m_i} \tag{95}$$

and now we are ready to define the `flux_coeff` so that `flux_coeff.size()` is $N-1$:

$$\texttt{flux\_coeff}[i-1] = f_i^* \tag{96}$$

Face values at the first node ($i = 0$)

Since the size of the `flux_coeff` array is $N-1$, and it should be multiplied element-wise by other N sized arrays (e.g., `lagrangian_coeff`) that are computed at the cell centers (specifically - the cell center of which the corresponding node is located at its left face since $i = j - 1/2$), we shall define the value of ρ^*, D^n, Δm at $i = 0$ by reflecting their values across the boundary, i.e., $f_0^* = f_1^*$.

3. We will also define the following numeric arrays for notation comfort. <u>For $j = 1, ..., N$:</u> Under the definitions:

$$\begin{aligned} \mathcal{L}_j^* &= \frac{\rho_j^*}{\Delta m_j} \\ B_j^n &= \frac{\chi c \sigma_j^n}{1 + A_j^n} \end{aligned} \tag{97}$$

The numeric arrays are:

$$\begin{aligned} \texttt{lagrangian\_coeff[j-1]} &= \mathcal{L}_j^* \\ \texttt{total\_coupling[j-1]} &= B_j^n \end{aligned} \tag{98}$$

4. To sum up, the coefficients are defined as follows. <u>For $j = 1, ..., N$ and $i = j - 1 = 0, ..., N-1$ (left face)</u>:

$$\begin{aligned} a_j^n &= -\mathcal{L}_j^* f_i^* \\ b_j^n &= \mathcal{L}_j^* (f_i^* + f_{i+1}^*) + \frac{1}{\Delta t} + B_j^n \\ c_j^n &= -\mathcal{L}_j^* f_{i+1}^* \\ d_j^n &= B_j^n [U_R]_j^n + \frac{1}{\Delta t} E_j^n \end{aligned} \tag{99}$$

and in code notation (after defining `a, b, c, d` arrays of `size = N`):

$$\begin{aligned}
\texttt{a[j - 1]} &= \mathcal{L}_j^* f_i^* \\
\texttt{b[j - 1]} &= \mathcal{L}_j^* (f_i^* + f_{i+1}^*) + \frac{1}{\Delta t} + B_j^n \\
\texttt{c[j - 1]} &= \mathcal{L}_j^* f_{i+1}^* \\
\texttt{d[j - 1]} &= B_j^n [U_R]_j^n + \frac{1}{\Delta t} E_j^n
\end{aligned} \tag{100}$$

Between code arrays and physical quantities

As stated at the beginning of this subsection, the physical quantities $\rho, \Delta m, D, \sigma$ are all given as arrays for the cell centers (i.e. $\rho_j =$ `rho[j - 1]`) for $j = 1, ..., N$ so assumptions such as `arr[j-1]` $= f(\rho, \Delta m, D, \sigma)$ just mean direct array assignment!

6 Appendix: Boundary Conditions in the Diffusion Approximation

In the P_1 diffusion approximation, the specific intensity $I(x, \mu)$ is not tracked for every angle. Instead, we use **Marshak Boundary Conditions** to bridge the gap between averaged quantities (energy density E and flux F) and the physical constraints at the boundaries.

6.1 General Physical Derivation

At a boundary interface, the net inward partial flux J_{in} must match the source drive. For a blackbody source at temperature T_{bath}, $J_{in} = \sigma_{sb} T_{bath}^4$. In the diffusion approximation, the partial fluxes are related to E and F by:

$$J_{in} = \frac{cE}{4} + \frac{F}{2}, \quad J_{out} = \frac{cE}{4} - \frac{F}{2} \tag{101}$$

At the origin ($x = 0$), the net flux into the material is F_0. By substituting $\sigma_{sb} T_{bath}^4 = \frac{c}{4} E_{bath}$, the Marshak condition becomes:

$$\frac{cE(0,t)}{4} + \frac{F_0(t)}{2} = \frac{c}{4} E_{bath}(t) \tag{102}$$

$$\boxed{F_0(t) = \frac{c}{2} E_{bath}(t) - \frac{c}{2} E(0,t)} \tag{103}$$

At the far boundary ($x = L$), we assume a vacuum where no radiation enters from the right ($J_{in,right} = 0$), leading to the free-streaming limit:

$$\frac{cE(L,t)}{4} - \frac{F_L(t)}{2} = 0 \implies \boxed{F_L = \frac{cE(L,t)}{2}} \tag{104}$$

6.2 Discretization in Lagrangian Coordinates

In our staggered grid, cell centers are indexed by $j = 1, \ldots, N$ and faces by $i = 0, \ldots, N$. The diffusion equation at the cell located between the faces i

and $i+1$ is:

$$\frac{E^{n+1}_{i+\frac{1}{2}} - E^{n}_{i+\frac{1}{2}}}{\Delta t} - \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i+1} D^n_{i+1} \frac{E^{n+1}_{i+\frac{3}{2}} - E^{n+1}_{i+\frac{1}{2}}}{\Delta m_{i+1}} \right)}^{-\hat{F}^{n+1}_{i+1}} + \frac{\rho^*_{i+\frac{1}{2}}}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left(\rho^n_{i} D^n_{i} \frac{E^{n+1}_{i+\frac{1}{2}} - E^{n+1}_{i-\frac{1}{2}}}{\Delta m_{i}} \right)}^{-\hat{F}^{n+1}_{i}} = \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} [U_R]^n_{i+\frac{1}{2}} - \frac{\chi c \sigma^n_{i+\frac{1}{2}}}{1 + A^n_{i+\frac{1}{2}}} E^{n+1}_{i+\frac{1}{2}} \tag{105}$$

Quick note about notation

I used the notation "$\hat{F}_1$" to indicate that the subscript denotes i and not j. For comfort indexing, in the notation "E_1" (without hat) the subscript denotes j (default notation prefers cell centers).

First Cell ($j = 1$)

For the first cell, we substitute the boundary flux $\hat{F}^{n+1}_0$ derived from the Marshak condition:

$$\frac{E^{n+1}_1 - E^n_1}{\Delta t} + \frac{\rho^*_1}{\Delta m_1} \left(\hat{F}^{n+1}_1 - \overbrace{\left[\frac{c}{2} E_{bath} - \frac{c}{2} E(0,t) \right]}^{\hat{F}^{n+1}_0} \right) = S^{n+1}_1 \tag{106}$$

where the source term is defined as:

$$S^{n+1}_1 = \frac{\chi c \sigma^n_1}{1 + A^n_1} [U_R]^n_1 - \frac{\chi c \sigma^n_1}{1 + A^n_1} E^{n+1}_1 \tag{107}$$

To close the system using only internal cell unknowns, we make the standard first-order spatial approximation that the radiation energy density at the boundary face is roughly equal to the energy of the first cell: $E(0,t) \approx E^{n+1}_1$. Substituting this and the internal flux $\hat{F}^{n+1}_1$ yields:

$$\frac{E^{n+1}_1 - E^n_1}{\Delta t} + \frac{\rho^*_1}{\Delta m_1} \left[\left(-\hat{\rho}^n_1 \hat{D}^n_1 \frac{E^{n+1}_2 - E^{n+1}_1}{\Delta \hat{m}_1} \right) - \frac{c}{2} E_{bath} + \frac{c}{2} E^{n+1}_1 \right] = \frac{\chi c \sigma^n_1}{1 + A^n_1} [U_R]^n_1 - \frac{\chi c \sigma^n_1}{1 + A^n_1} E^{n+1}_1 \tag{108}$$

Notice that the surface emission term has become $\frac{\rho_1^* c}{2\Delta m_1} E_1^{n+1}$. This naturally moves to the LHS of the equation to maintain the implicit scheme stability. Let us define this Marshak boundary cooling coefficient C_1:

$$C_1 = \frac{c\rho_1^*}{2\Delta m_1} \tag{109}$$

Grouping terms for E_1^{n+1} and E_2^{n+1} yields the modified tridiagonal system:

$$\begin{aligned} &\left(\frac{1}{\Delta t} + \frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} + C_1 + \frac{\chi c \sigma_1^n}{1 + A_1^n} \right) E_1^{n+1} \\ &- \left(\frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} \right) E_2^{n+1} = \\ &\frac{E_1^n}{\Delta t} + C_1 E_{bath} + \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n \end{aligned} \tag{110}$$

where $E_{bath} = aT_{bath}^4(t)$.

This gives us the modified tridiagonal coefficients for the first cell:

$$\boxed{\tilde{b}_1^n = \frac{1}{\Delta t} + \frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} + C_1 + \frac{\chi c \sigma_1^n}{1 + A_1^n}} \tag{111}$$

$$\boxed{\tilde{c}_1^n = -\frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1}} \tag{112}$$

$$\boxed{\tilde{d}_1^n = \frac{E_1^n}{\Delta t} + C_1 E_{bath} + \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n} \tag{113}$$

Relation to the base internal coefficients

We can connect these modified coefficients to the base internal coefficients a_1^n, b_1^n, d_1^n derived earlier. Recall that a_1^n represents the left-face diffusion contribution, which is $-\frac{\rho_1^* \hat{\rho}_0^n \hat{D}_0^n}{\Delta m_1 \Delta \hat{m}_0}$, and b_1^n contains $-a_1^n$. The Marshak condition essentially replaces this left-face diffusion with the Marshak cooling/drive terms.
Therefore, the modified coefficients can be elegantly constructed directly from the base coefficients:

$$\boxed{\tilde{b}_1^n = b_1^n + a_1^n + C_1} \tag{114}$$

$$\boxed{\tilde{a}_1^n = 0} \tag{115}$$

$$\boxed{\tilde{d}_1^n = d_1^n + C_1 E_{bath}} \tag{116}$$

This avoids recalculating the shared terms (like the right-face flux or the source terms) and is exactly how it is implemented numerically.

Last Cell ($i = N - \frac{1}{2}, j = N$): This is not done in our simulation because the Marshak waves don't reach the edge boundary in our simulation setup.

$$\frac{E_N^{n+1} - E_N^n}{\Delta t} + \frac{\rho_N^*}{\Delta m_N} \left(\frac{c E_N^{n+1}}{2} - F_{N-1}^{n+1} \right) = S_N^{n+1} \tag{117}$$

This modifies the final diagonal element:

$$\boxed{\tilde{b}_N^n = b_N^n + \frac{c \rho_N^*}{2 \Delta m_N}} \tag{118}$$

Missing implementation details

I've not included here the implementation of the calculation of $T_{bath}(t)$ from using Menahem's subsonic solver. What it actually does is use the semi-analytic solver to draw the

7 Calculation of $T_{bath}(t)$ from Menahem's subsonic solver

When solving Marshak BC problems, one gives $T_{bath}(t)$ rather than $T_{surface}(t)$ as the BC. In our **building of test cases**, we would like to calculate $T_{bath}(t)$ that when used in the simulation via Marshak BC is **consistent with some other Dirichlet BC** solution.

Derivation of driving radiation bath temperature

At the boundary ($m = 0$), the net radiation energy flux entering the system is governed by the physical Marshak boundary condition:

$$\frac{F(0,t)}{2} = \sigma_{\rm sb}\left[T^4_{\rm bath}(t) - T^4_s(t)\right] \tag{119}$$

Given the power-law surface material temperature boundary condition, $T_s(t) = T_0 t^\tau$, this yields:

$$T_{\rm bath}(t) = \left[T_0^4 t^{4\tau} + \frac{F(0,t)}{2\sigma_{\rm sb}}\right]^{1/4} \tag{120}$$

From the subsonic similarity formulation (Eq. 638), the physical flux boundary value is matched to the self-similar dimensional radiation energy flux at the origin ($\xi = 0$), $F(0,t) = s(0,t)$:

$$F(0,t) = A^{a+(q+1)a_1+na_3} B^{1+b+(q+1)b_1+nb_3} t^{c+(q+1)c_1+nc_3} S(0) \tag{121}$$

By fully substituting the structural expressions for the powers computed in the code:

$$\begin{aligned}
n &= \frac{4+\alpha}{\beta} \\
k &= 1-\mu \\
q &= (1-\mu)\left(\frac{4+\alpha}{\beta}\right) + \lambda - 1 \\
\text{denom} &= \frac{1}{4+2\lambda-4\mu}
\end{aligned}$$

We write the explicit algebraic components for the exponents $a, b, c,$

a_1, b_1, c_1, and a_3, b_3, c_3:

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{4 + 2\lambda - 4\mu} \tag{122}$$

$$a_1 = \frac{-2(4+\alpha-2\beta)}{4 + 2\lambda - 4\mu} \tag{123}$$

$$a_3 = \frac{-2\left[\mu(4+\alpha) - \beta\lambda - 4 - \alpha\right]}{4 + 2\lambda - 4\mu} \tag{124}$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} \tag{125}$$

$$b_1 = \frac{-2}{4 + 2\lambda - 4\mu} \tag{126}$$

$$b_3 = \frac{-2(\mu - 1)}{4 + 2\lambda - 4\mu} \tag{127}$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau\left[2(\beta - \alpha - 4) - \beta\lambda + \mu(4+\alpha)\right]}{4 + 2\lambda - 4\mu} \tag{128}$$

$$c_1 = \frac{-2\left[\tau(4+\alpha-2\beta) - 1\right]}{4 + 2\lambda - 4\mu} \tag{129}$$

$$c_3 = \frac{-2\left[\tau\left(\mu(4+\alpha) - \beta\lambda - 4 - \alpha\right) + 1 - \mu\right]}{4 + 2\lambda - 4\mu} \tag{130}$$

Now, substituting the definitions of the dimensional constants A and B:

$$A = T_b((\gamma-1)f)^{1/\beta}, \quad B = \frac{16\sigma_{\mathrm{sb}} g}{3\beta((\gamma-1)f)^n} \tag{131}$$

The cumulative explicit time-independent pre-factor is:

$$\left[T_b((\gamma-1)f)^{1/\beta}\right]^{a+(q+1)a_1+na_3} \left[\frac{16\sigma_{\mathrm{sb}} g}{3\beta((\gamma-1)f)^n}\right]^{1+b+(q+1)b_1+nb_3} S(0) \tag{132}$$

Substituting this complete expansion back into the Marshak expression, we isolate $T_0 t^\tau$ outside the bracket to reveal the explicit non-equilibrium boundary correction term:

$$T_{\mathrm{bath}}(t) = T_0 t^\tau \left[1 + \mathcal{M}_{\mathrm{explicit}} \cdot t^{\Delta\eta}\right]^{1/4} \tag{133}$$

Where the fully expanded constant scalar $\mathcal{M}_{\mathrm{explicit}}$ is defined by:

$$\mathcal{M}_{\mathrm{explicit}} = \frac{S(0)}{\sigma_{\mathrm{sb}} T_0^4} \left[T_b((\gamma-1)f)^{1/\beta}\right]^{a+(q+1)a_1+na_3} \left[\frac{16\sigma_{\mathrm{sb}} g}{3\beta((\gamma-1)f)^n}\right]^{1+b+(q+1)b_1+nb_3} \tag{134}$$

And the explicit net power law deviation parameter $\Delta\eta$ is:

$$\begin{aligned}
\Delta\eta &= c + (q+1)c_1 + nc_3 - 4\tau && (135) \\
&= \frac{1}{4+2\lambda-4\mu}\Big\{2\left[(4+\alpha)(1-\beta^{-1}) - 1 - \lambda\right] \\
&\quad + \tau\left[2\beta - \lambda(4+\alpha) - 4\alpha - 18\right]\Big\} - 4\tau && (136)
\end{aligned}$$

Under strict thermodynamic self-similarity where the internal drive completely tracks the external source, $\Delta\eta = 0$, meaning $c+(q+1)c_1+nc_3 = 4\tau$. In this configuration, the temporal term cancels from the bracket, delivering a constant scaling parameter over the entire timeline:

$$T_{\text{bath}}(t) = T_0\left[1 + \frac{S(0)}{\sigma_{\text{sb}}T_0^4}\left[T_b((\gamma-1)f)^{1/\beta}\right]^{a+(q+1)a_1+na_3}\left[\frac{16\sigma_{\text{sb}}g}{3\beta((\gamma-1)f)^n}\right]^{1+b+(q+1)b_1+nb_3}\right]^{1/4} t^{\tau} \tag{137}$$